

- **Automated Trading Agent:** Create an RL-based trading bot for stocks or cryptocurrency. The agent makes buy/sell decisions and receives rewards based on the profit or loss of those decisions. Over time, it aims to maximize returns by discovering trading strategies that work well in the market.
- **Traffic Signal Optimization:** Implement an RL agent to control traffic lights in a simulated city grid. The agent gets positive rewards when traffic flow improves (shorter waits, reduced congestion) and negative rewards for long queues or accidents. By learning from these signals, it optimizes the timing of lights to reduce overall traffic jam times.

## Real-World Applications

- **Finance**
  - Fraud detection (**Classification, Anomaly Detection**)
  - Stock price prediction (**Regression, Time-Series Forecasting, Reinforcement Learning**)
- **Healthcare**
  - Disease diagnosis (**Classification**)
  - Medical imaging analysis (**Classification, Clustering, Dimensionality Reduction**)
- **Marketing**
  - Recommendation systems (**Clustering, Classification, Association Rule Learning**)
  - Customer segmentation (**Clustering**)
- **Manufacturing**
  - Predictive maintenance (**Regression, Anomaly Detection**)
  - Quality control (**Classification, Anomaly Detection**)

## Conclusion

Machine learning helps computers learn from data and make predictions without explicit programming. Understanding supervised, unsupervised, and reinforcement learning provides a foundation for applying ML across various industries.